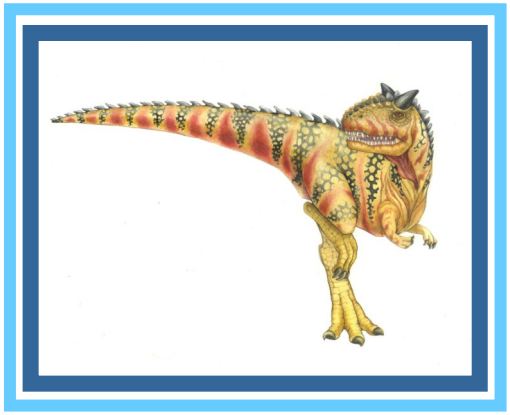
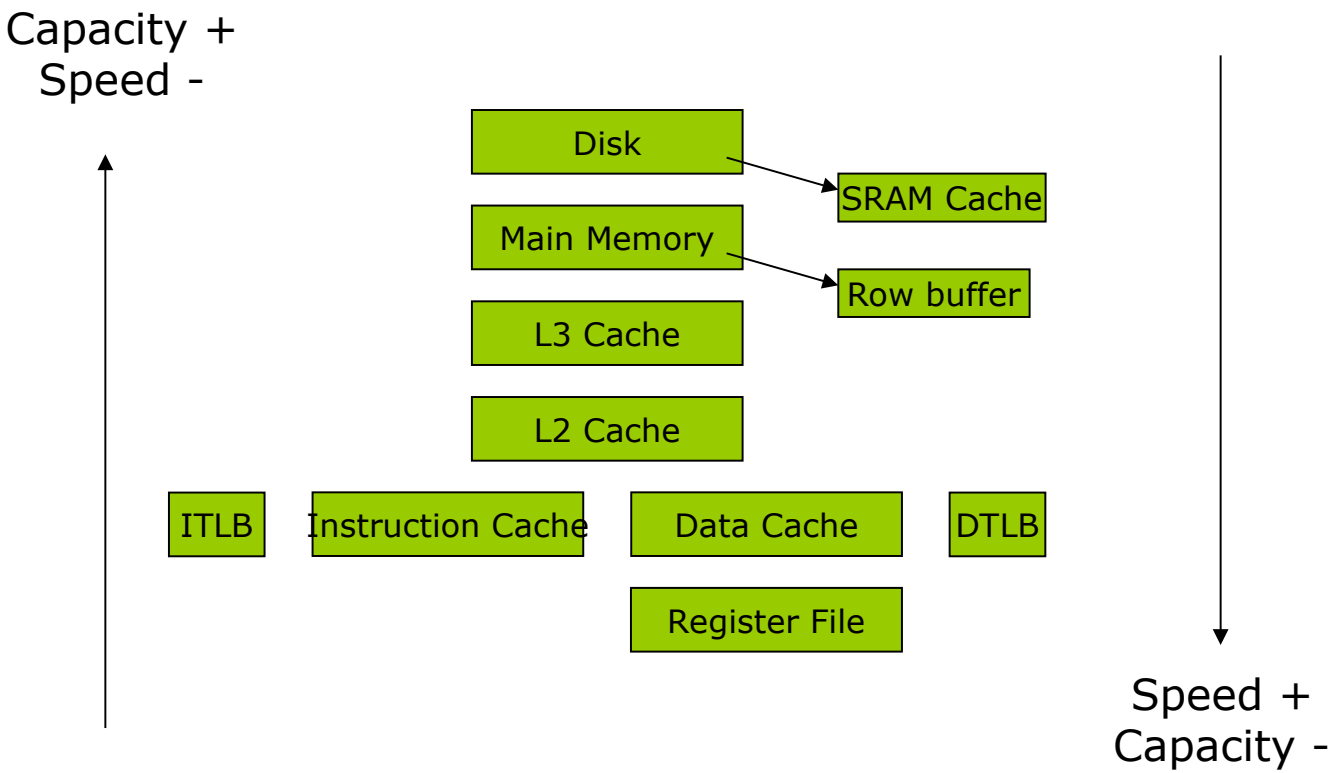
Chapter 12: Cache

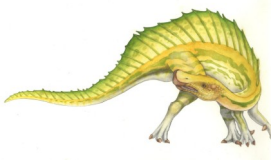


Storage Hierarchy and Locality



Memory Latency is *Long*

- 60-100ns not totally uncommon
- Quick back-of-the-envelope calculation:
  - 2GHz CPU
  - → 0.5ns / cycle
  - 100ns memory → 200 cycle memory latency!
- Solution: Caches



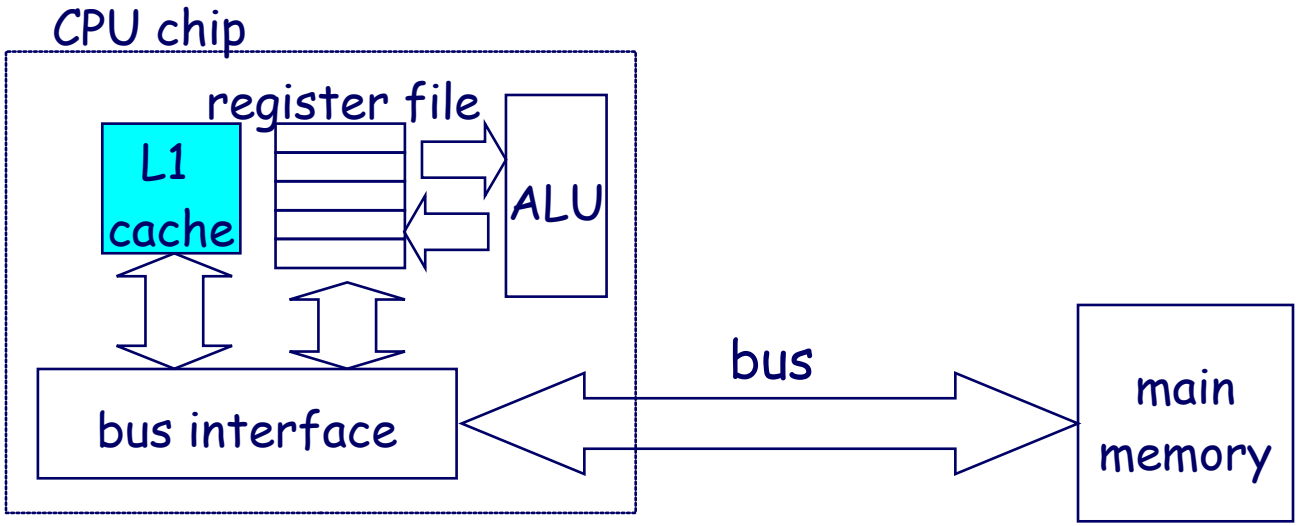
Hardware cache memories

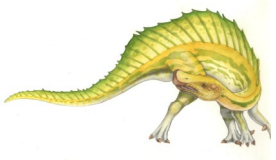
Cache memories are small, fast SRAM-based memories managed automatically in hardware

- Hold frequently accessed blocks of main memory

CPU looks first for data in L1, then in main memory

Typical system structure:

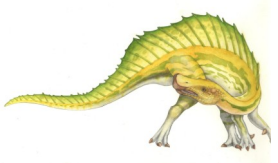




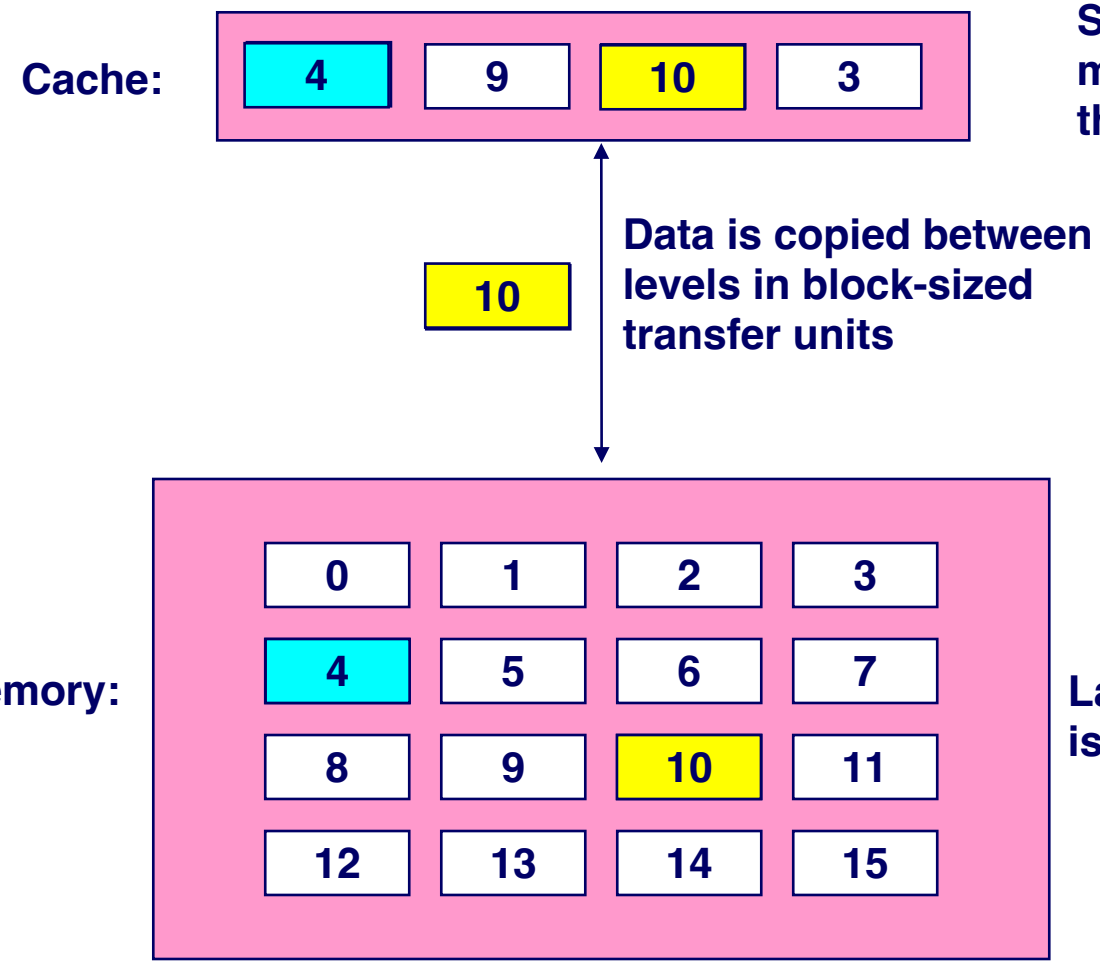
Cache Basics

- Fast (but small) memory close to processor
- When data referenced
  - If in cache, use cache instead of memory
  - If not in cache, bring into cache (actually, bring entire **block** of data, too)
  - Maybe have to kick something else out to do it!

Key: Optimize the *average* memory access latency



General cache mechanics



Smaller, faster, more expensive memory caches a subset of the blocks

Larger, slower, cheaper memory is partitioned into “blocks”

From lecture-9.ppt

Cache Performance Metrics

Miss Rate

- Fraction of memory references not found in cache (misses / accesses)
  - 1 – hit rate ☺
- Typical numbers (in percentages):
  - 3-10% for L1
  - can be quite small (e.g., < 1%) for L2, depending on size, etc.

Hit Time

- Time to deliver a line in the cache to the processor
  - includes time to determine whether the line is in the cache
- Typical numbers:
  - 1-2 clock cycle for L1
  - 5-20 clock cycles for L2

Miss Penalty

- Additional time required because of a miss
  - typically 50-200 cycles for main memory (Trend: increasing!)

Lets think about those numbers

Huge difference between a hit and a miss

- 100X, if just L1 and main memory

Would you believe 99% hits is twice as good as 97%?

- Consider these numbers:

cache hit time of 1 cycle

miss penalty of 100 cycles

So, average access time is:

97% hits: 0.97 * 1 cycle + 0.03 * 100 cycles = 4 cycles

99% hits: 0.99 * 1 cycle + 0.01 * 100 cycles = 2 cycles

Average/effective access time =

hit_rate*cache_latency + (1-hit_rate)*memory_latency

In Class Practice

- A CPU is equipped with a cache; Accessing a word takes 20 clock cycles if the data is in not in the cache and 5 clock cycles if the data is in the cache.
- (1) What is the effective memory access time in clock cycles if the hit ratio is 90%.
- (2) What should be the hit ratio if the effective memory access time is 8 clock cycles.

- (1) 6.5
- (2) 80%

Key Issues Related To Cache

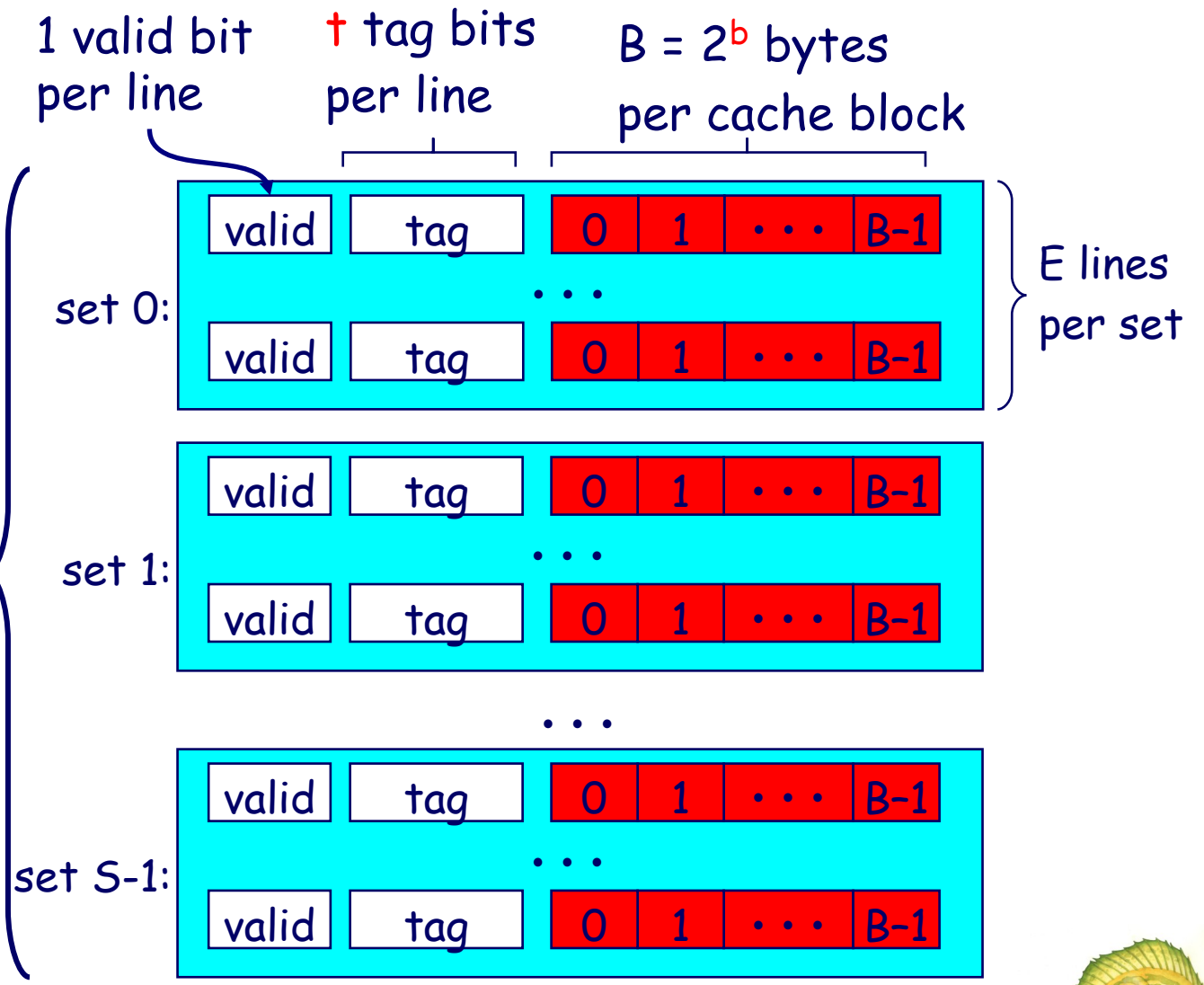
- Important decisions
  - Structure: what is the general organization of cache?
  - Identification: how do we find a block in cache?
  - Placement: where in the cache can a block go?
  - Replacement: what to kick out to make room in cache?
    - LRU can also be applied here

General Organization of a Cache

Cache is an array of sets

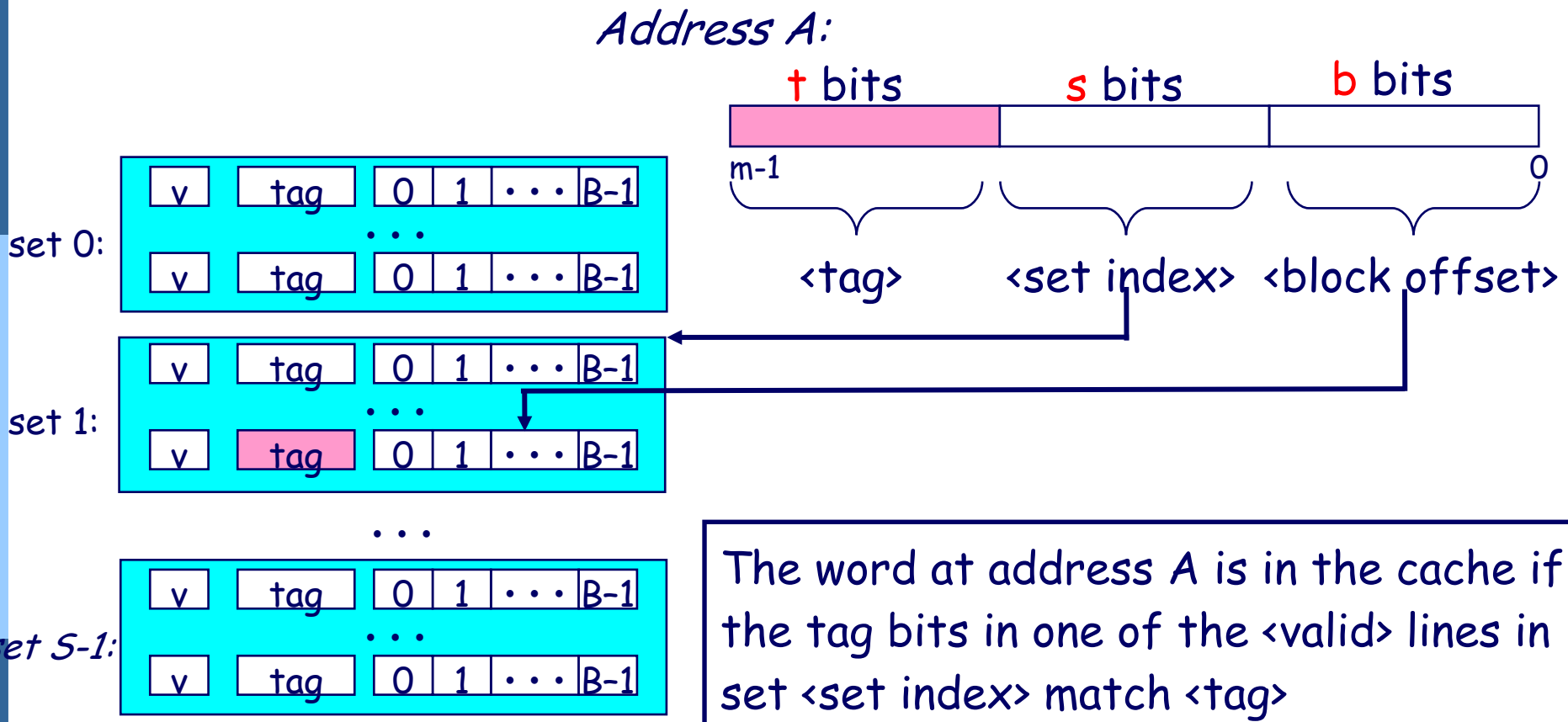
Each set contains one or more lines

Each line holds a block of data



Cache size: C = B x E x S data bytes

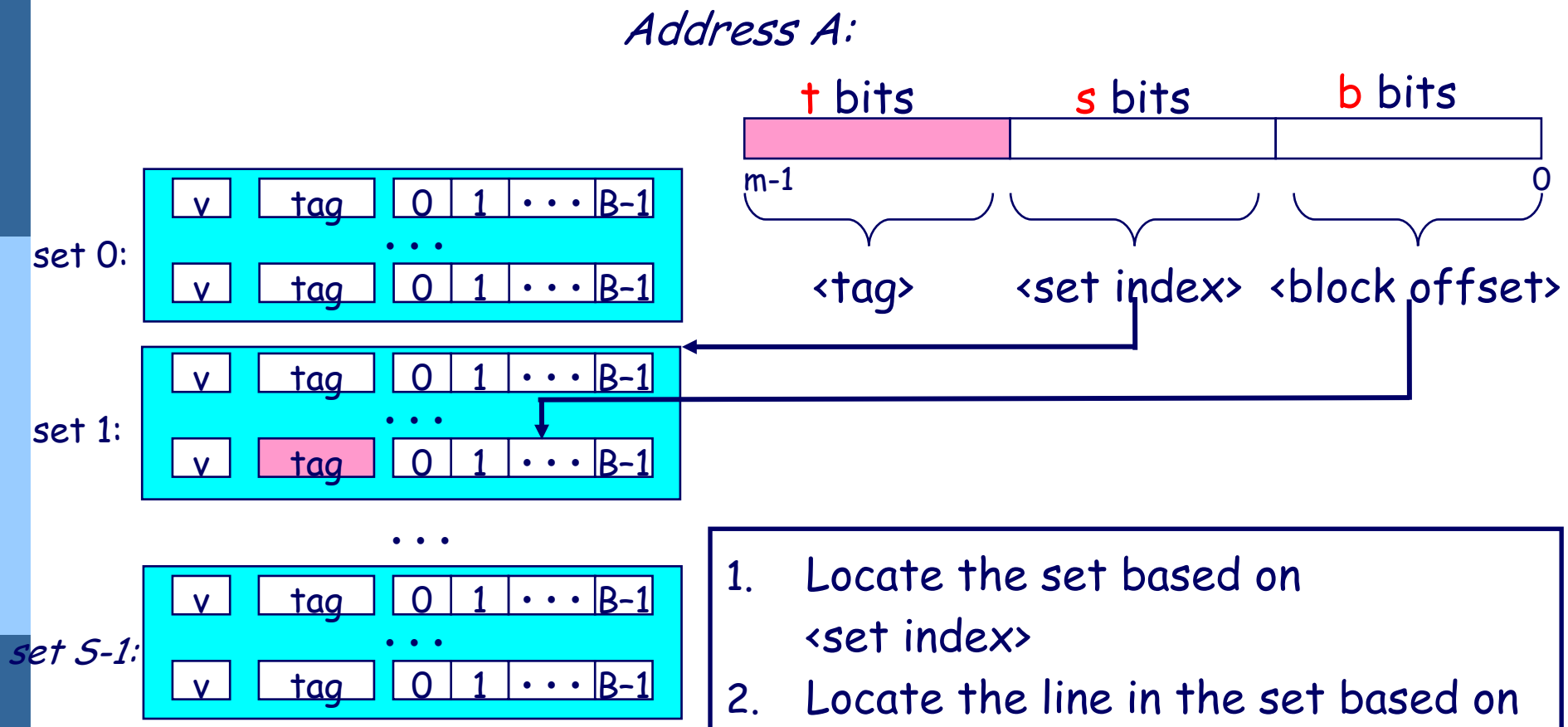
Addressing Caches



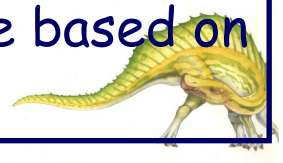
The word at address A is in the cache if the tag bits in one of the <valid> lines in set <set index> match <tag>

The word contents begin at offset <block offset> bytes from the beginning of the block

Addressing Caches

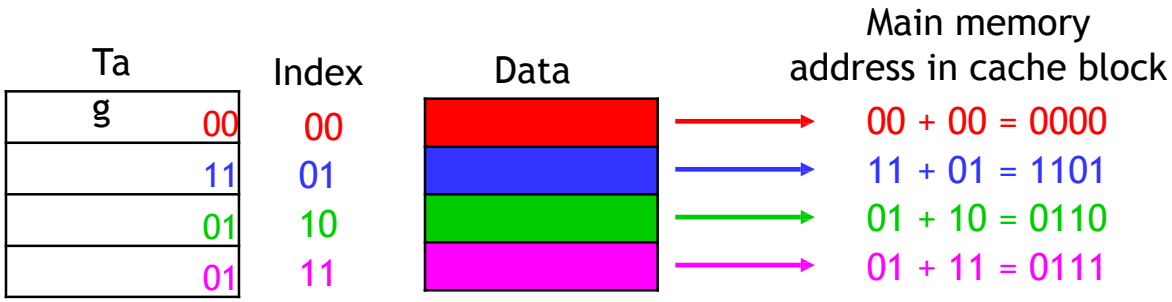


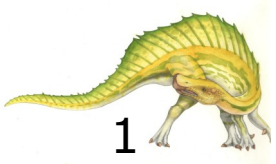
1. Locate the set based on <set index>
2. Locate the line in the set based on <tag>
3. Check that the line is valid
4. Locate the data in the line based on <block offset>



Figuring out what's in the cache

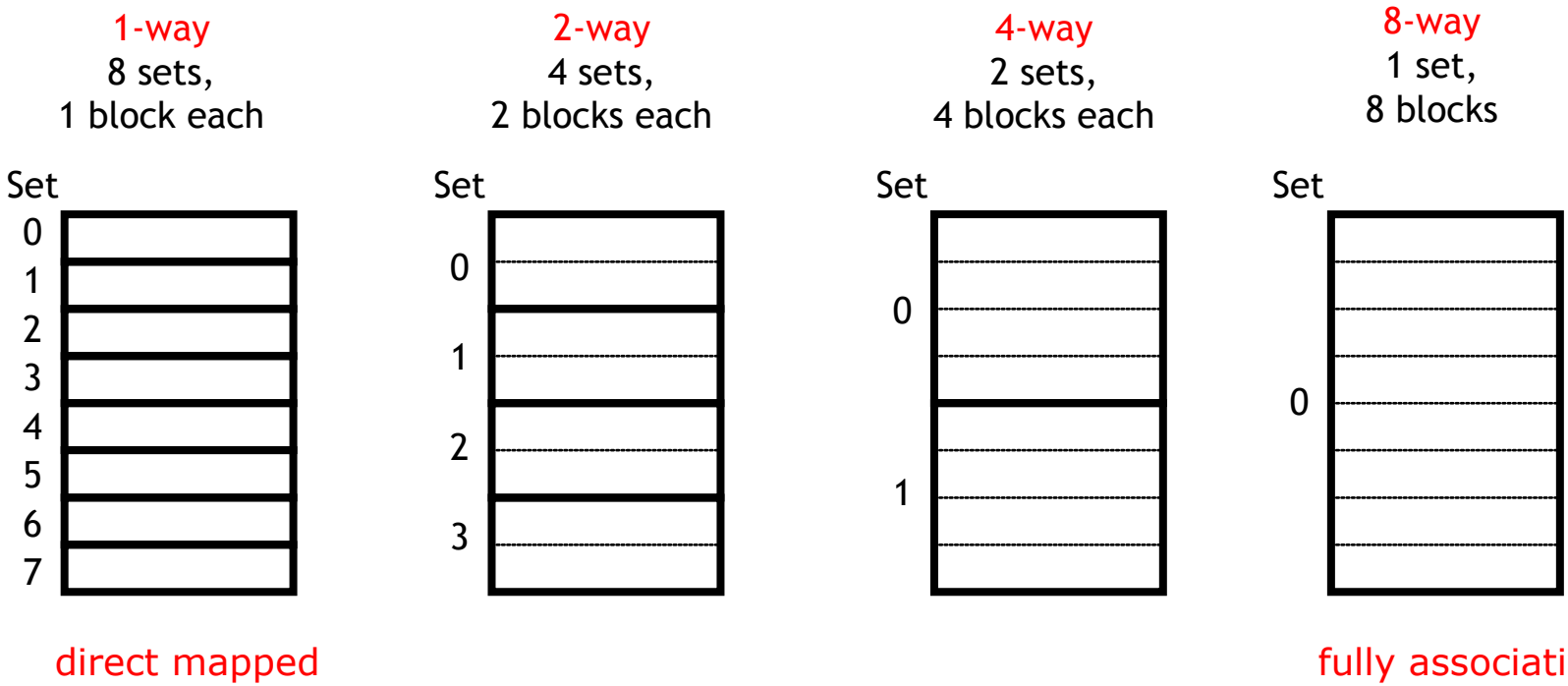
- Set Index in the middle.
- Now we can tell exactly which addresses of main memory are stored in the cache, by concatenating the cache block tags with the block indices.





Set associative caches are a general idea

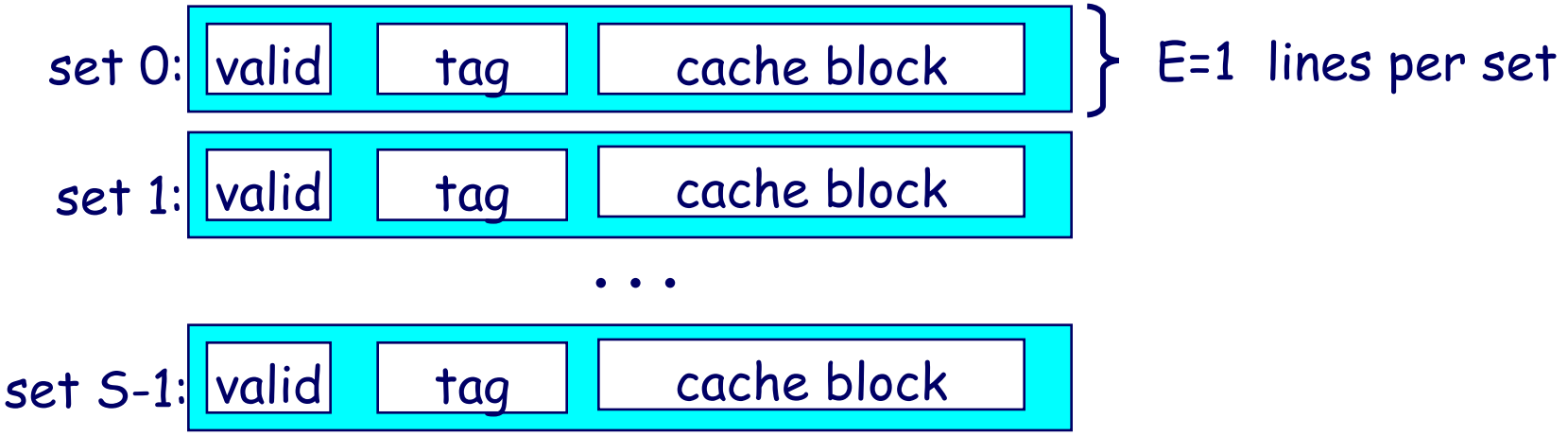
- By now you may have noticed the 1-way set associative cache is the same as a direct-mapped cache.
- Similarly, if a cache has 2^k blocks, a 2^k-way set associative cache would be the same as a fully-associative cache.



Example: Direct-Mapped Cache

Simplest kind of cache, easy to build, and fastest mapping
(only 1 tag compare required per access)

Characterized by exactly one line per set.



Direct Mapping

- Which line to store the information?
  - Line ID = Block ID mod number of lines
- How many lines are there in a cache?
  - number of lines = cache size / words per line
  - E.g., cache size 2048 words, 16 words per line, there are 128 lines
  - Line bits = log2(number of lines)
  - Cache size: 2^(line bits + word bits)
- How many tags are there?
  - number of tags = number of blocks / number of lines (or memory size / cache size)
  - Tag bits = log2(number of tags)
  - Memory size: 2^(tag bits + line bits + word bits)